

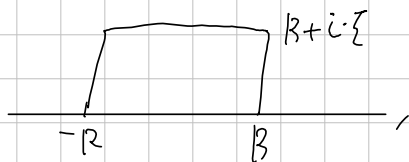
✓ Evaluate

$$\int_{-\infty}^{\infty} e^{-\pi x^2} \cdot e^{-2\pi i x \zeta} dx = e^{-\pi \zeta^2} \quad (\zeta > 0)$$

Proof. Let $f(z) = e^{-\pi z^2}$. Then, it is entire.

Consider four contour: $C_1: t \quad (-R \leq t \leq R)$,
 $C_2: R + it \quad (0 \leq t \leq \zeta)$
 $C_3: t + i\zeta \quad (-R \leq t \leq R)$
 $C_4: -R + it \quad (0 \leq t \leq \zeta)$

Then, the composition $C = C_1 \cdot C_2 \cdot (-C_3) \cdot (-C_4)$ is described as



and, by the Cauchy's theorem, $\int_C f = 0$. Now,

$$\int_{C_1} f = \int_{-R}^R e^{-\pi x^2} dx,$$

$$\int_{C_2} f = \int_0^{\zeta} e^{-\pi(R+it)^2} \cdot i dt = i \int_0^{\zeta} e^{-\pi(R^2 - t^2 + 2iRt)} \cdot dt$$

$$\text{and so } \left| \int_{C_2} f \right| = \left| \int_0^{\zeta} e^{\pi \cdot (t^2 - R^2)} dt \right| \leq \left| \int_0^{\zeta} e^{\pi t^2} dt \right| \cdot e^{-\pi R^2} \rightarrow 0 \text{ as } R \rightarrow \infty.$$

Similarly $\left| \int_{C_4} f \right| \rightarrow 0$ as $R \rightarrow \infty$. And,

$$\begin{aligned} \int_{C_3} f &= \int_{-R}^R e^{-\pi(t+i\zeta)^2} dt = \int_{-R}^R e^{-\pi(t^2 - \zeta^2 + 2it\zeta)} dt \\ &= e^{\pi \zeta^2} \cdot \int_{-R}^R e^{-\pi t^2} \cdot e^{-2\pi it\zeta} dt \end{aligned}$$

Thus, as $R \rightarrow \infty$,

$$0 = \int_{-\infty}^{\infty} e^{-\pi x^2} dx - e^{\pi \zeta^2} \int_{-\infty}^{\infty} e^{-\pi t^2} \cdot e^{-2\pi it\zeta} dt. \quad \square$$

